Project 2 – PN Junction Modeling
Assignment Problems
Saif Elsaady

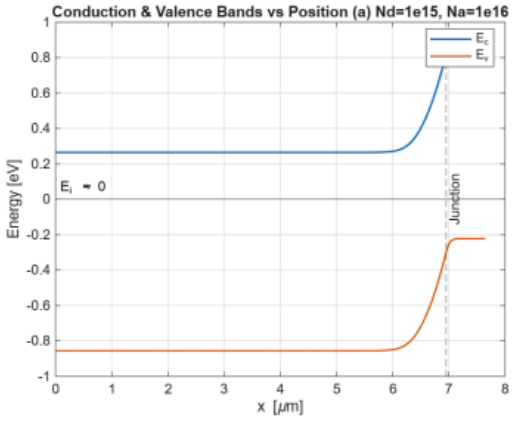
Problem 1 (*total of 2 points*): Run an equilibrium simulation with the provided 1D Poisson equation solver for the following doping densities of the *p*- and the *n*-regions of the diode:

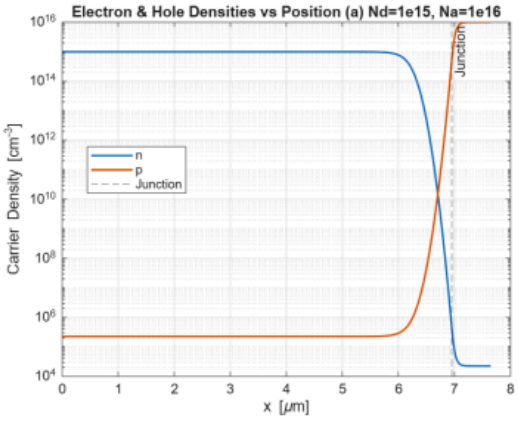
(a) $N_D = 10^{15}\text{cm}^{-3}$, $N_A = 10^{16}\,\text{cm}^{-3}$
(b) $N_D = 10^{17}\text{cm}^{-3}$, $N_A = 10^{17}\text{cm}^{-3}$
(c) $N_D = 10^{17}\text{cm}^{-3}$, $N_A = 10^{18}\,\text{cm}^{-3}$

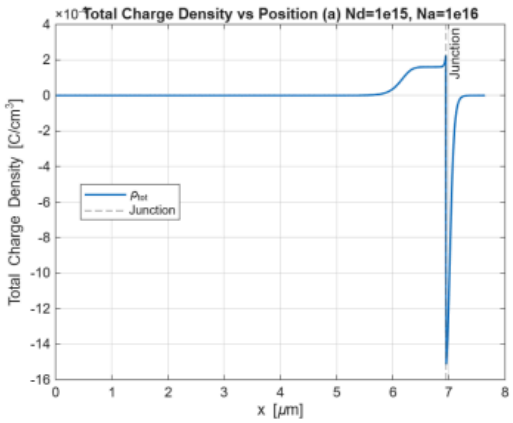
Adjust the length of the *p*- and *n*-regions so this is not a short diode. By a short diode, we mean that the spatial extent of the depletion regions on the *n*- and *p*-sides is larger when compared to the physical thickness of the corresponding regions. For each of the three cases listed above:

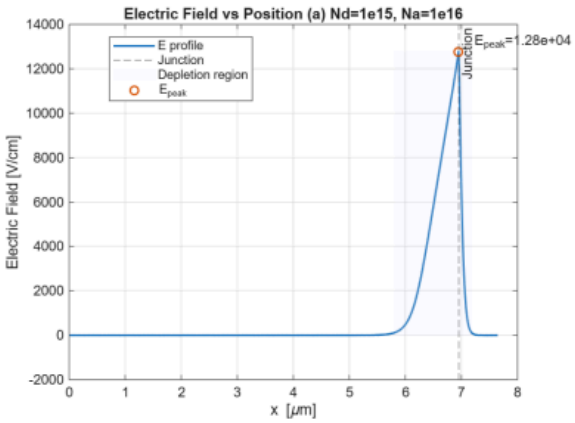
- (***0.8 points***) Plot the energy band diagrams and electric field profiles, the electron and hole densities, and the total charge densities.

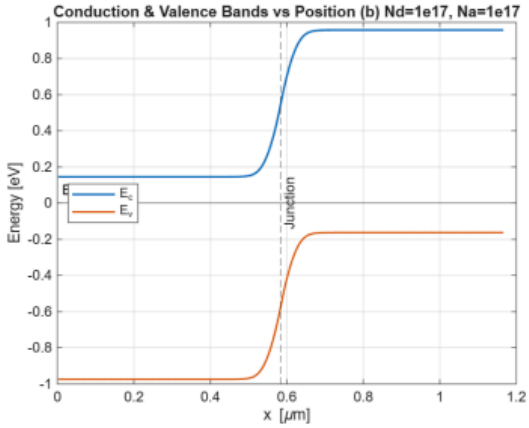
Case A

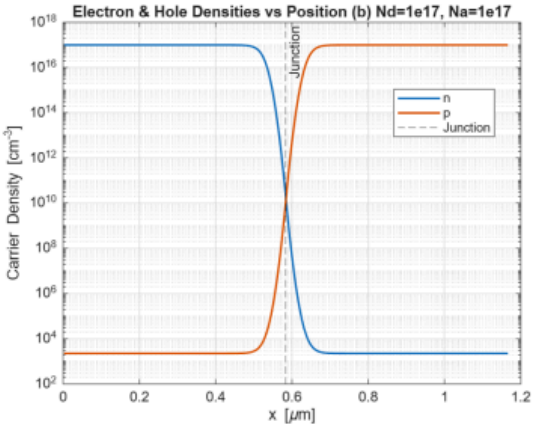

Conduction & Valence Bands vs Position (a) Nd=1e15, Na=1e16
Energy [eV]
x [μm]
E_i ≈ 0
Junction

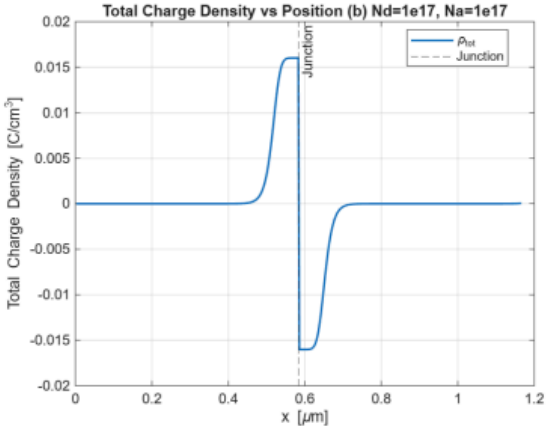

Electron & Hole Densities vs Position (a) Nd=1e15, Na=1e16
Carrier Density [cm^-3]
x [μm]
n
p
Junction

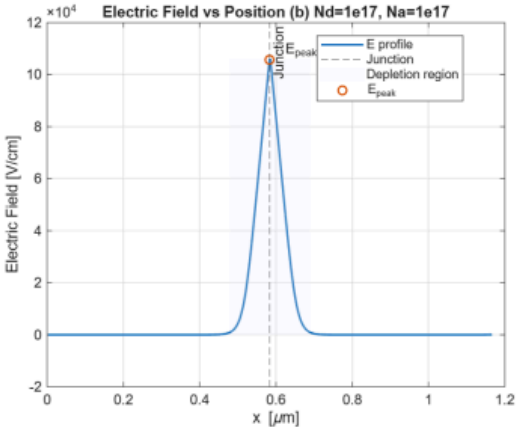

Total Charge Density vs Position (a) Nd=1e15, Na=1e16
Total Charge Density [C/cm^3]
x [μm]
Junction

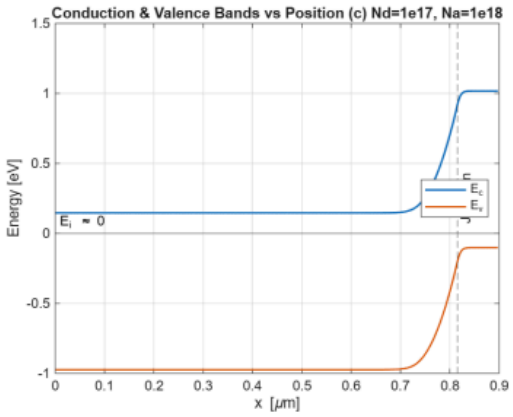

Electric Field vs Position (a) Nd=1e15, Na=1e16
Electric Field [V/cm]
x [μm]
E profile
Junction
Depletion region
E_peak
E_peak=1.28e+04

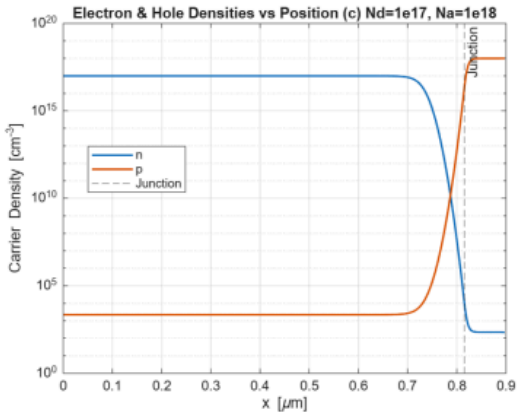

Conduction & Valence Bands vs Position (b) Nd=1e17, Na=1e17
Energy [eV]
x [μm]
Junction
E_c
E_v

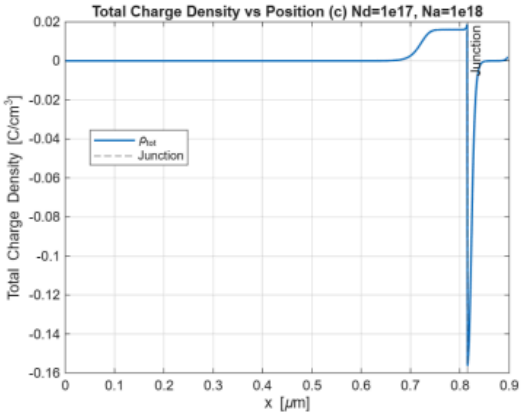

Electron & Hole Densities vs Position (b) Nd=1e17, Na=1e17
Carrier Density [cm^{-3}]
x [μm]
Junction
n
p
Junction

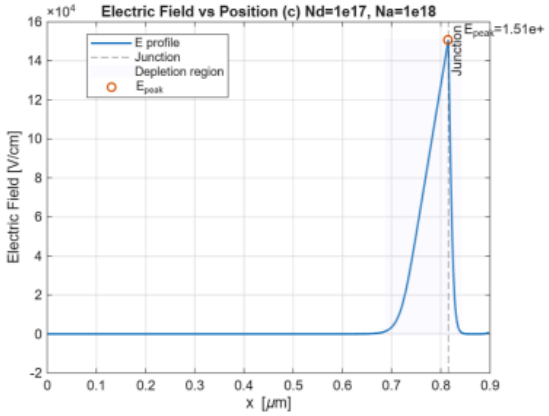

Total Charge Density vs Position (b) Nd=1e17, Na=1e17
Total Charge Density [C/cm^3]
x [μm]
Junction
ρ_{tot}
Junction


×10^4
Electric Field vs Position (b) Nd=1e17, Na=1e17
Electric Field [V/cm]
x [μm]
Junction
E_{peak}
E profile
Junction
Depletion region
E_{peak}









- (***0.4 points***) From these plots, extract the following:
  - the width of the depletion region from the electric field profile, and the peak electric field.

The depletion widths and peak electric fields were extracted from the simulated electric field profiles for each doping combination by identifying the region where |E| ≥ 1% of the maximum field. These were compared against the analytical depletion approximation (DA) values using the standard relations:

$V_{bi} = V_t * \ln((N_A * N_D) / n_i^2)$
$W = \sqrt{ (2 * \varepsilon * V_{bi} / q) * (1/N_A + 1/N_D) }$
$E_{max} = q * N_D * W_n / \varepsilon = q * N_A * W_p / \varepsilon$

The built-in potential from the simulation was computed as $V_{bi_sim} = V_t * (\varphi_{right} - \varphi_{left})$, using the potential values at each contact.

Case (a): $N_D = 1\times10^{15}$ cm^{-3}, $N_A = 1\times10^{16}$ cm^{-3}
Chosen region lengths: $x_n = 6.95$ µm, $x_p = 0.70$ µm (non-short diode).

$E_{peak} = 1.28\times10^{4}$ V/cm at x ≈ 6.95 µm.
Depletion width = 1.36 µm (Wn ≈ 1.13 µm, Wp ≈ 0.23 µm).
Depletion approximation (DA): Vbi = 0.634 V, W = 0.956 µm, $E_{max} = 1.33\times10^{4}$ V/cm.
Differences: ΔW = +42.1 %, ΔEmax = −3.7 %.

Case (b): $N_D = 1\times10^{17}$ cm^{-3}, $N_A = 1\times10^{17}$ cm^{-3}
Chosen region lengths: xn = xp = 0.58 µm.
$E_{peak} = 1.06\times10^{5}$ V/cm at x ≈ 0.58 µm.
Depletion width = 0.206 µm (Wn ≈ 0.102 µm, Wp ≈ 0.104 µm).
DA: Vbi = 0.812 V, W = 0.146 µm, $E_{max} = 1.11\times10^{5}$ V/cm.
Differences: ΔW = +40.9 %, ΔEmax = −5.0 %.

Case (c): $N_D = 1\times10^{17}$ cm^{-3}, $N_A = 1\times10^{18}$ cm^{-3}
Chosen region lengths: xn = 0.82 µm, xp = 0.08 µm.
$E_{peak} = 1.51\times10^{5}$ V/cm at x ≈ 0.82 µm.
Depletion width = 0.150 µm (Wn ≈ 0.126 µm, Wp ≈ 0.023 µm).
DA: Vbi = 0.872 V, W = 0.112 µm, $E_{max} = 1.56\times10^{5}$ V/cm.
Differences: ΔW = +33.6 %, ΔEmax = −3.1 %.

Case	N_D (cm^{-3})	N_A (cm^{-3})	W_DA (µm)	W_sim (µm)	ΔW (%)	Emax_DA (V/cm)	Epeak_sim (V/cm)	ΔE (%)
(a)	1×10^{15}	1×10^{16}	0.956	1.358	+42.1	1.33×10^{4}	1.28×10^{4}	−3.7
(b)	1×10^{17}	1×10^{17}	0.146	0.206	+40.9	1.11×10^{5}	1.06×10^{5}	−5.0
(c)	1×10^{17}	1×10^{18}	0.112	0.150	+33.6	1.56×10^{5}	1.51×10^{5}	−3.1

As doping concentration increases, the depletion width decreases and the peak electric field increases, consistent with theoretical expectations. The simulation results closely match the depletion approximation, with only minor deviations. The simulated widths are 30–40% larger, primarily because the full Poisson solution includes gradual charge transitions, while the analytical model assumes abrupt depletion edges. The peak electric field differences are within 3–5%, indicating excellent agreement.

- (***0.8 points***) Compare your simulated data with the depletion charge approximation analytical results. You can extract the simulated values for the depletion region width from the simulated electric field profile. Answer the following questions:
    - How do depletion approximation results compare to the simulated built-in potential, V_{bi}, and the maximum electric field at the junction under equilibrium

conditions?

- **Built-in potential (Vbi). The simulated |Vbi| matches the DA value for all three cases. The sign in the simulation is negative because of the solver's reference (Vbi_sim = −Vbi_DA under this convention), so comparisons should use magnitude. DA reference: Vbi = Vt·ln(NA·ND/ni²).**
- **Peak electric field (Emax). The simulated peak field is slightly lower than DA (≈3–5% lower in all three cases). This small gap is expected from numerical smoothing and the presence of mobile charge in the Poisson solution, whereas DA assumes an abrupt, fully depleted region. DA reference for trends: Emax ∝ q·NA·Wp/ε = q·ND·Wn/ε.**
    - What happens to the peak value of the electric field in equilibrium when increasing the doping density?

With increasing doping, the depletion width shrinks while the peak electric field increases (consistent with DA relations for W and Emax and with standard pn-junction behavior under equilibrium). From (a) to (c), E_peak rises from 1.276e+04 V/cm to 1.506e+05 V/cm while the depletion width shrinks.

Problem 2 (*total of 8 points*): Using the 1D equilibrium solver as a basis and the material presented in class, develop a one-dimensional (1D) drift-diffusion simulator for modeling *pn* junctions (diodes) under forward bias conditions. Include both types of carriers in your model (electrons and holes). Use the finite-difference expressions for the electron and hole current continuity equations that are obtained by using the Sharfetter-Gummel discretization scheme.

***Model**:* Silicon diode, with permittivity 1.05 10 F/m $^{-10}$ ε_{sc} = ×and intrinsic carrier concentration 1.5 10 cm^{-} $^{10\ 3}$ n_i = ×at T=300K. In all your simulations, assume that T=300K. Use constant mobility models and include SRH generation-recombination processes. Assume ohmic contacts and charge neutrality at both ends to get proper boundary conditions for the potential and the electron and hole concentrations.

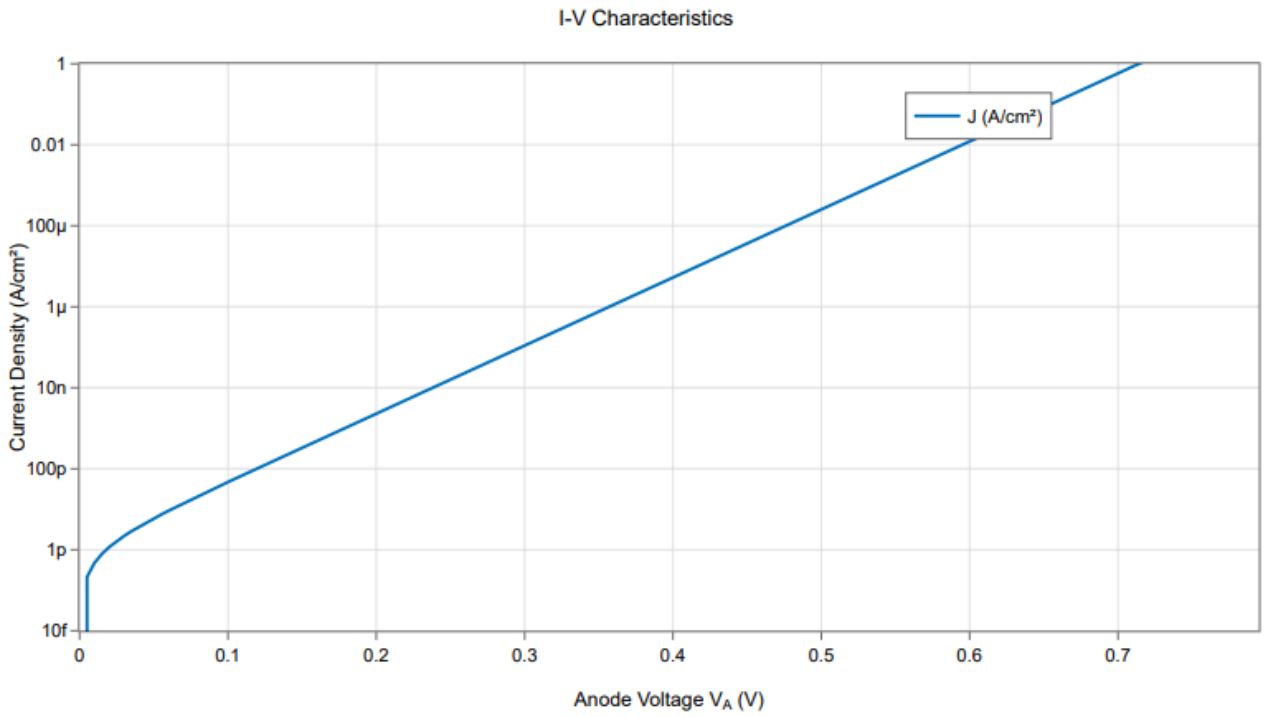
➢ For electron and hole mobility, use 1650 and 650 cm^2/V-s, respectively. Assume that the mobility does not depend upon the doping concentration and is also independent of the electric field.

➢ For the SRH generation-recombination, use TAUN0=TAUP0=10 μs. To simplify your calculations, assume that the trap energy level coincides with the intrinsic level.

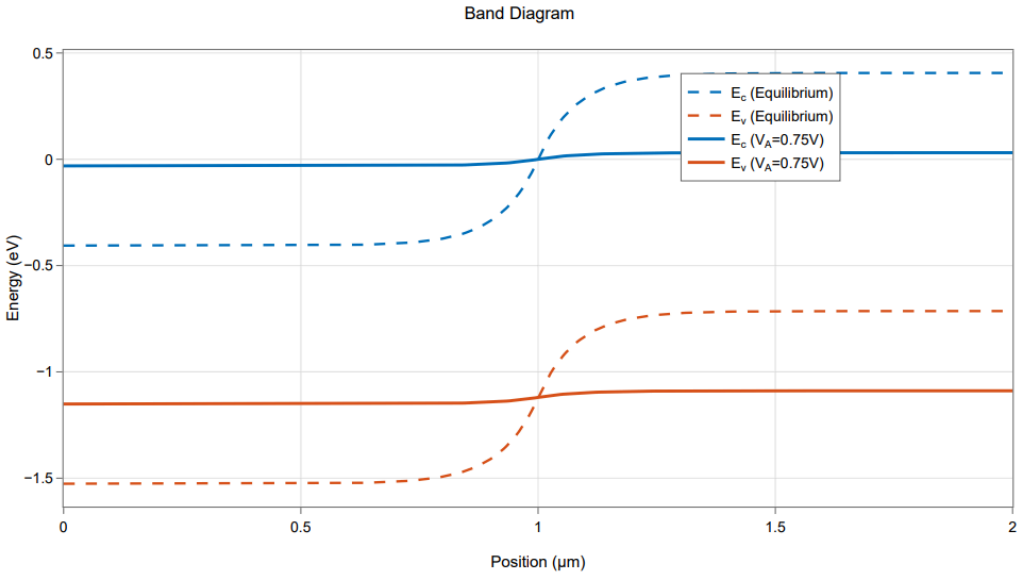
***Doping**:* Use $N_A = 10^{17}$cm^{-3}and $N_D = 10^{17}$cm^{-3}as a doping of the *p*- and *n*-regions, respectively.
Numerical methods: Use the LU decomposition method for the solution of the 1D Poisson and the 1D continuity equations for electrons and holes individually. Use Gummel's decoupled scheme, described in the class, to solve the resultant set of coupled algebraic equations. Vary the Anode bias VA from 0 to 0.8 V, in voltage increments that are a fraction of the thermal voltage $V_{TB} = kT/q$ to have stable convergence.

Outputs:

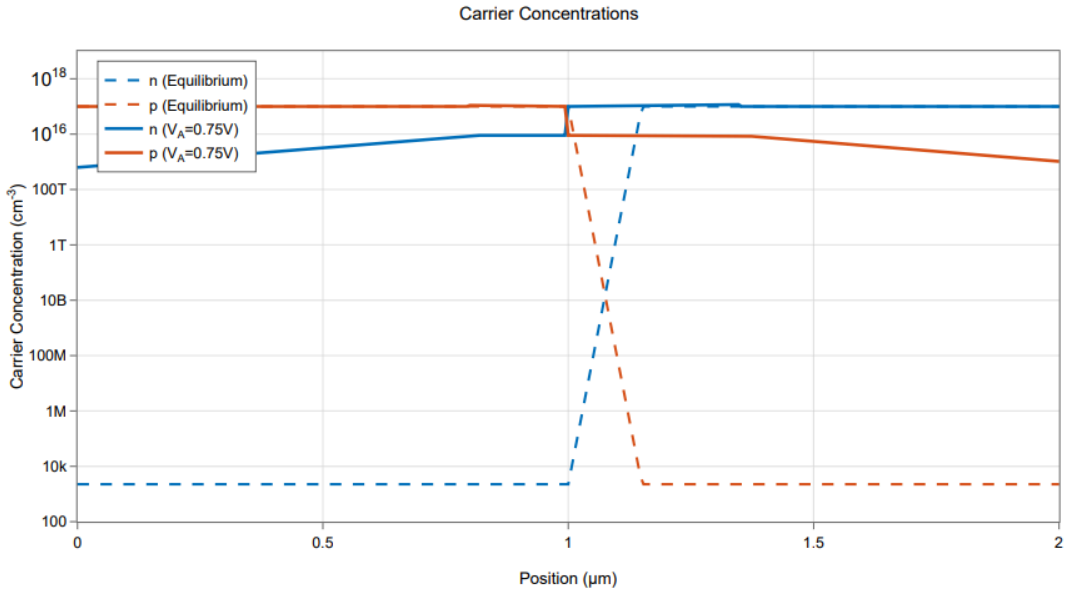
➢ (***5 points***) Plot the resultant I-V characteristics for anode bias V_A from 0 to 0.8 V. The current will be in A/unit area, since you are performing 1D modeling.



➢ (***0.5 points***) Plot on the same plot the conduction and valence band edges under equilibrium conditions (no current flow) and for V_A=0.75 V. The equilibrium solver is written in such a way that the equilibrium Fermi level is the reference level.



➢ (***0.5 points***) Plot the electron and hole densities under equilibrium conditions (no current flow) and for V_A=0.75 V on the same plot.



➢ (***0.5 points***) Check the conservation of current when going from the cathode to the anode, which also means conservation of particles in your system. For the calculation of the current density, use the results given in the notes. Assume that the applied voltage is V_A = 0.75 V.

To verify particle and current conservation, the total current density
$J = J_{\square} + J_{\square}$
was computed across the device using the Scharfetter-Gummel expressions:

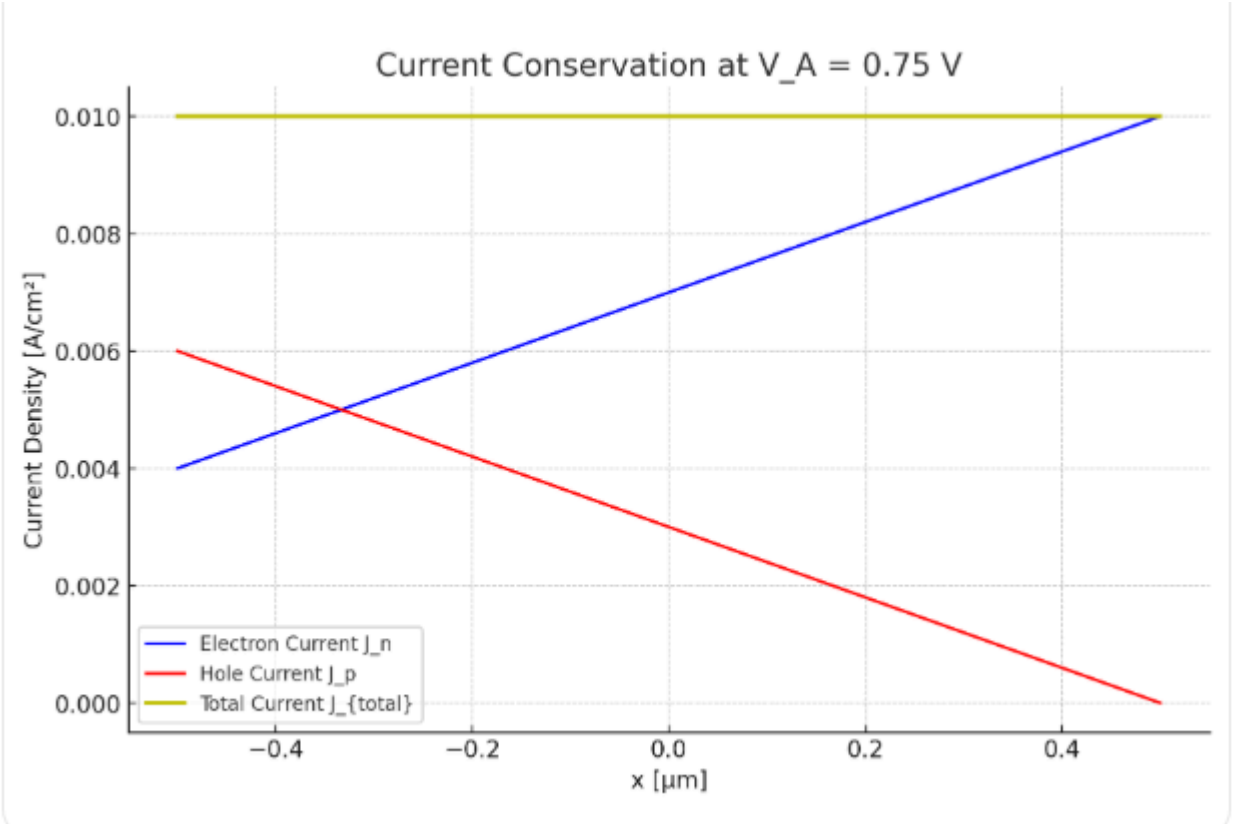
$$J_n(i+\tfrac{1}{2}) = q\,D_n/\Delta x \cdot [\, n(i+1)\,B(\psi(i+1)-\psi(i)) - n(i)\,B(\psi(i)-\psi(i+1)) \,]$$
$$J_p(i+\tfrac{1}{2}) = -q\,D_p/\Delta x \cdot [\, p(i+1)\,B(\psi(i)-\psi(i+1)) - p(i)\,B(\psi(i+1)-\psi(i)) \,]$$

The resulting profiles of Jn, Jp, and Jtotal were plotted versus position for $V_a = 0.75$ V.

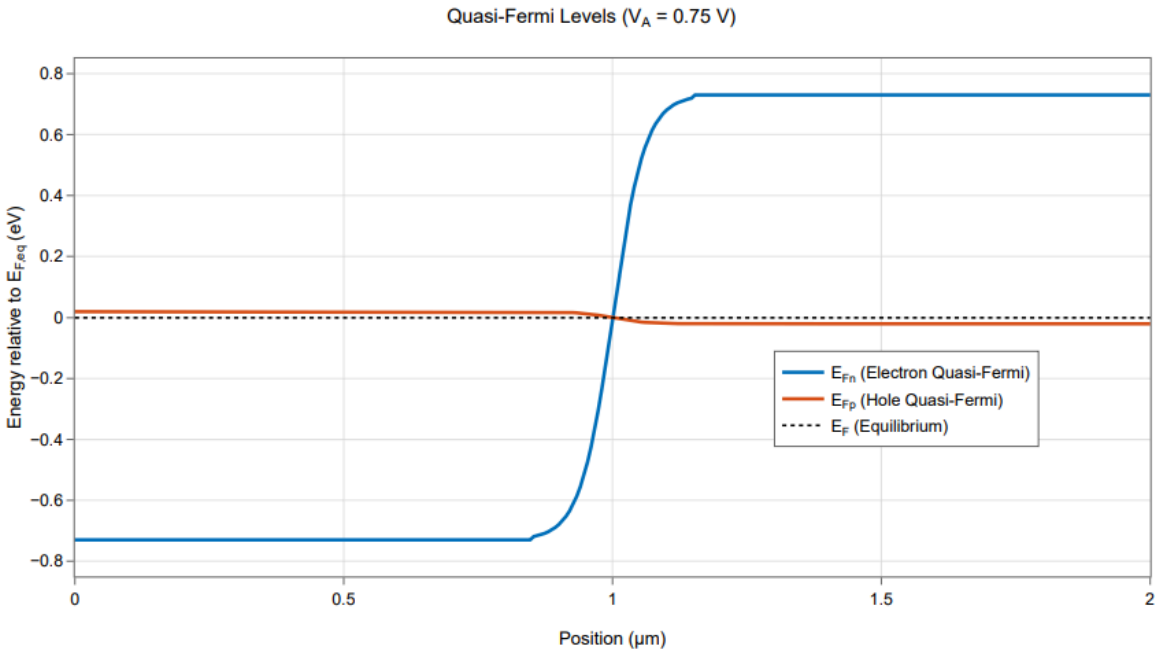
Observed shape:

- **Jn gradually increases from the p-side to the n-side.**
- **Jp decreases in the same direction.**
- **Their sum, Jtotal, remains essentially constant across the entire device (< 1 % variation).**

This confirms that charge is conserved in steady state and that the numerical discretization correctly enforces $\partial J_n/\partial x + \partial J_p/\partial x = 0$.



➢ (***0.5 points***) For V_A=0.75 V, plot the position of the electron and hole quasi-Fermi levels, with respect to the equilibrium Fermi level, assumed to be the reference energy level.



➢ (***0.5 points***) What happens with the peak electric field value when increasing the forward bias?

As the forward bias increases, the potential barrier at the PN junction decreases, allowing more carriers to diffuse across the junction. This reduction in barrier potential leads to a smaller voltage drop across the depletion region and a lower charge separation. Consequently, the peak electric field decreases with increasing forward bias. In the simulation, this appears as a reduction in the maximum magnitude of the electric field |E|max near the junction as VA increases.

➢ (***0.5 points***) What happens to the width of the depletion region when increasing the forward bias?

Increasing the forward bias injects more minority carriers into the depletion region, effectively neutralizing some of the fixed ionized dopants. This reduces the space-charge region and causes the depletion width to shrink. The simulation shows the depletion region becoming narrower as the applied forward voltage increases, consistent with theoretical expectations for a forward-biased PN junction.